

Trends in diet and Alzheimer's disease during the nutrition transition in Japan and developing countries.

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BACKGROUND: Alzheimer's disease (AD) rates in Japan and developing countries have risen rapidly in recent years. Researchers have associated factors such as the Western diet, obesity, alcohol consumption, and smoking with risk of AD. OBJECTIVE: This paper evaluates whether the dietary transition might explain the rising trend of AD prevalence in Japan and in developing countries, evaluating other factors when possible. METHODS: This study used two approaches to see whether dietary or other changes could explain AD trends in Japan and developing countries. One approach involved comparing trends of AD in Japan with changes in national dietary supply factors, alcohol consumption, and lung cancer mortality rates from zero to 25 years before the prevalence data. The second compared AD prevalence values for eight developing countries with dietary supply factors from zero to 25 years before the prevalence data. RESULTS: For Japan, alcohol consumption, animal product, meat and rice supply, and lung cancer rates correlated highly with AD prevalence data, with the strongest correlation for a lag of 15-25 years. In the eight-country study, total energy and animal fat correlated highly with AD prevalence data, with a lag of 15-20 years. Mechanisms to explain the findings include increased obesity for the eight countries, and increases in cholesterol, saturated fat, and iron from increases in animal products and meat supply for Japan. CONCLUSION: Evidently AD rates will continue rising in non-Western countries for some time unless we address major risk factors involving diet, obesity, and smoking.